



Ignite IT Performance™

Performance Tuning Made Easy

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Who Am I?

CONFIO
SOFTWARE
Ignite IT Performance™

Microsoft
CERTIFIED
Database Administrator



ignite8
CONFIO®



follow me
@SQLRockstar



<http://thomaslarock.com>



I Wrote A Book!

<http://dbasurvivor.com/>



Do You Have Value?



What Are You Measured By?





Performance Tuning is Hard!

- Takes time





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- 3-5 Day Classes are typical





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Performance Tuning is Hard!

- Takes time
- 3-5 Day Classes are typical
- Cannot be everything to everyone
- Requires expertise in many areas
- Low Priority
- Where do you begin?



What Do People Crave?





DMAIC For Everyone!

- DMAIC
 - Define
 - Measure
 - Analyze
 - Improve
 - Control



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What Can DMAIC Do For Me?

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What Can DMAIC Do For Me?

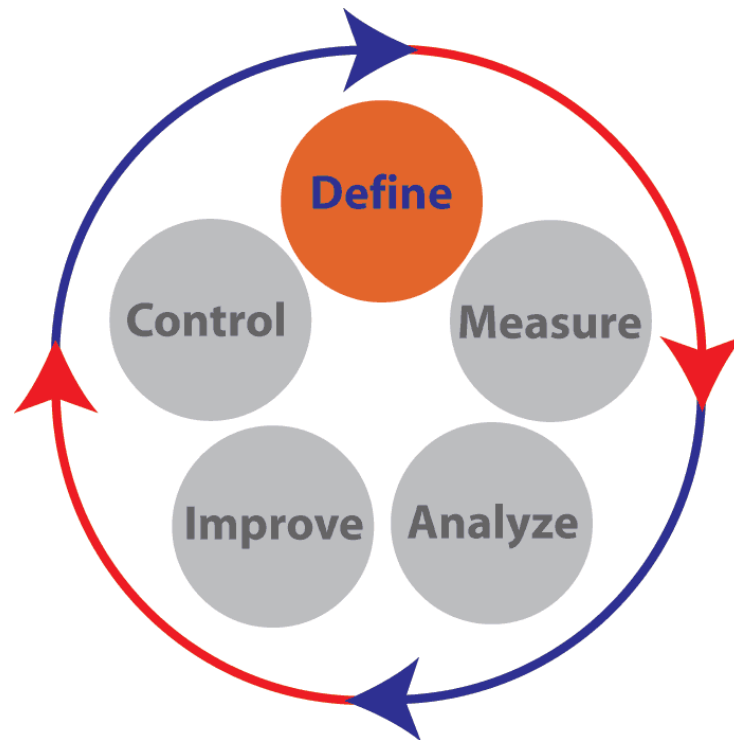
- Help you locate bottlenecks
- Help you resolve bottlenecks
- Help you become efficient
- Help you to demonstrate *your value*

What Can DMAIC Do For Me?

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- How are issues defined?
- What are the challenges you face when trying to define issues?



- Common sources for issues:
 - Benchmarks
 - User experience
 - Business requirements keep changing

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 - Benchmarks
 - User experience
 - Business requirements keep changing
- Common challenges:
 - User perception
 - OLTP vs. OLAP
 - Business requirements keep changing

- What are you measuring?
- How do you measure?



- Common measures:

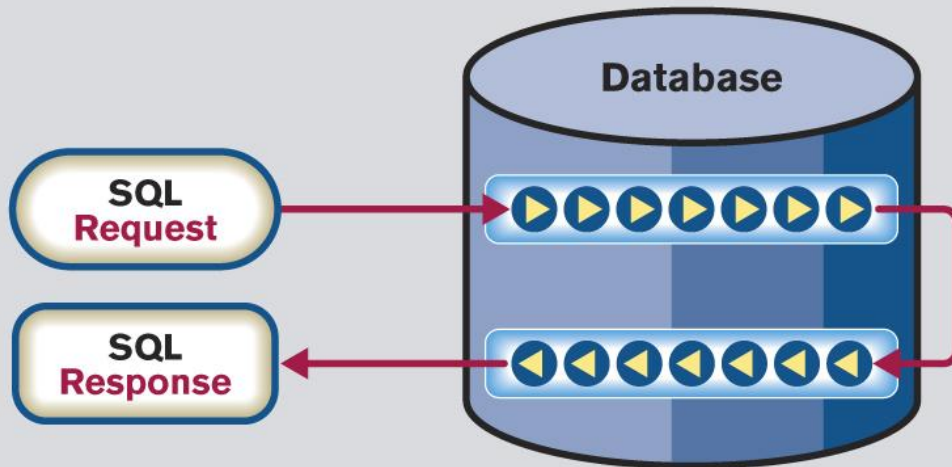
- Common measures:
 - Memory
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 - Disk I/O
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 - Memory
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 - Network
- How are metrics gathered?
 - SQL Trace
 - DMVs
 - 3rd party tools

} “Roll Your Own”



Focus on End User Response Time



Identify Wait-Time at every step and rank bottlenecks by user impact.

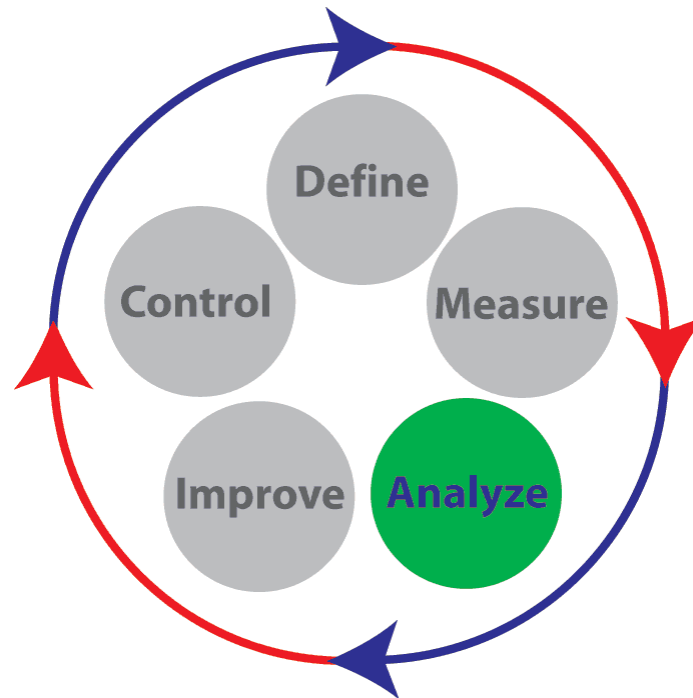
- Understand the total time a Query spends in Database
- Measure time while Query executes
- SQL Server helps by providing Wait Types

Common Wait Types:

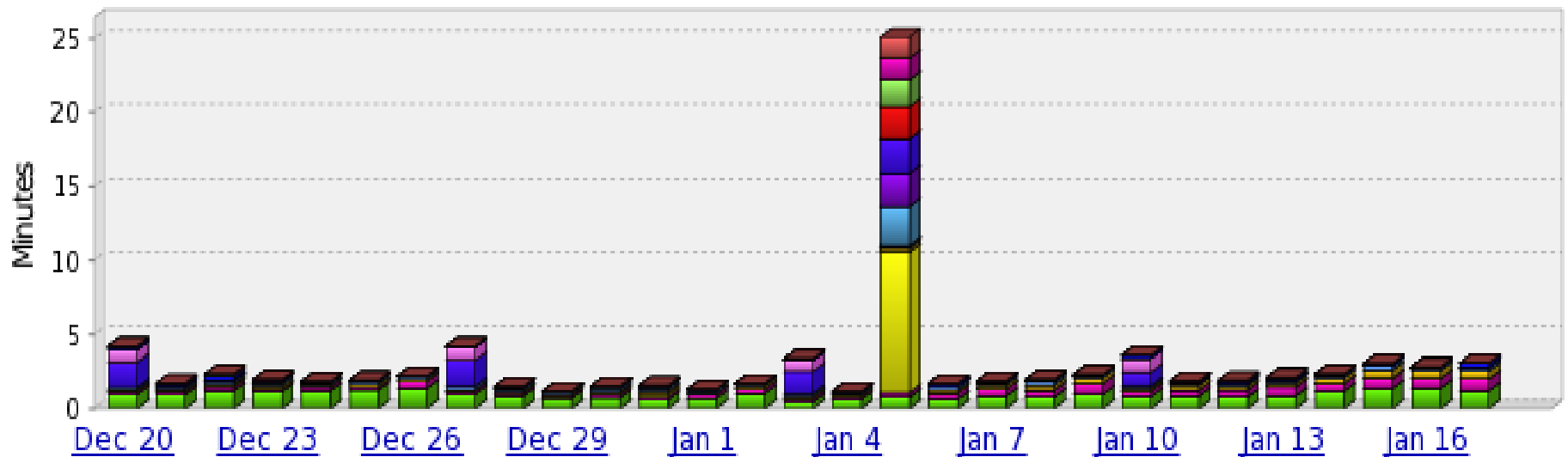
- ASYNC_NETWORK_IO
- CXPACKET
- WRITELOG
- PAGEIOLATCH_x
- LOCK_x
- LATCH_x

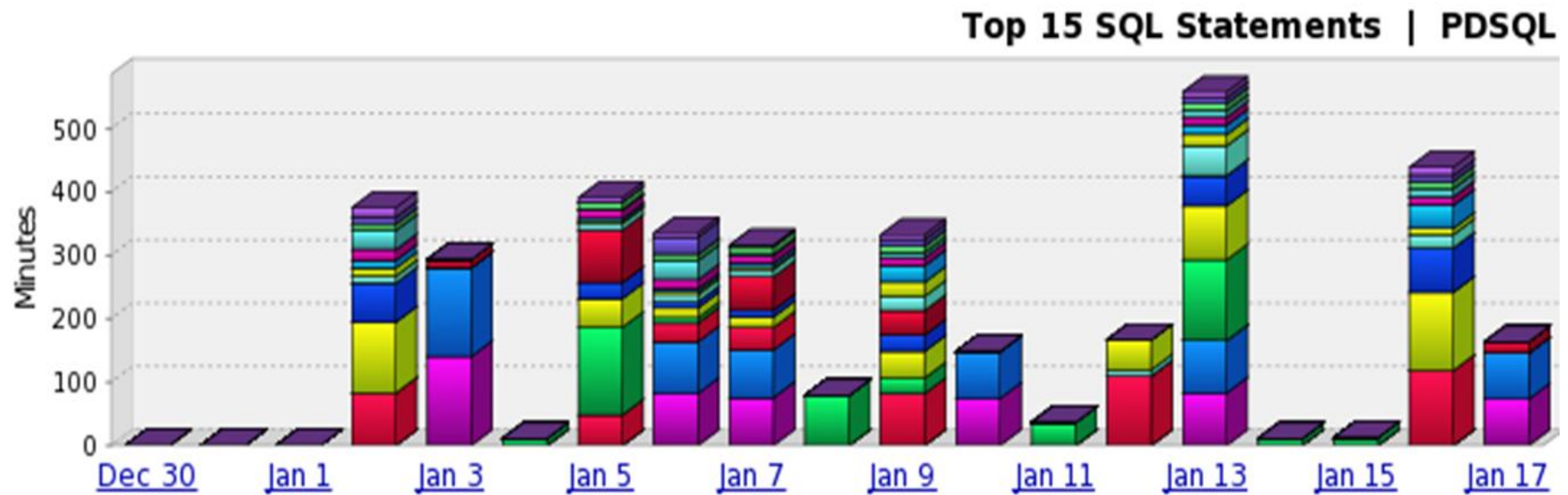
<http://technet.microsoft.com/en-us/library/cc966413.aspx>

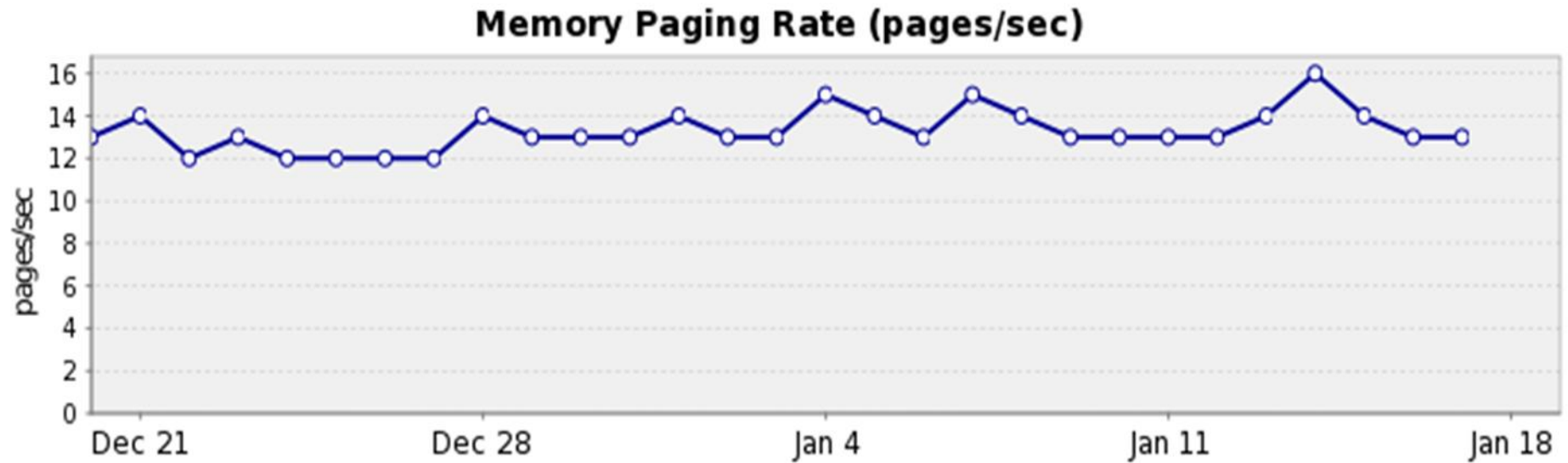
- Is it a problem?
- Experience is valued here the most
- How do you analyze?



Top 15 SQL Statements | DWERPPESSQL05D\DWPPERP







- How do you make improvements?
- Where do you look to make changes first?





DMAIC: Improve

- Which scenario is worse?
- SQL Statement 1
 - Executed 100 times
 - Caused 10 minutes of wait time for end user
 - Waited 90% of time on "PAGEIOLATCH_SH"



- Which scenario is worse?
- SQL Statement 1
 - Executed 100 times
 - Caused 10 minutes of wait time for end user
 - Waited 90% of time on "PAGEIOLATCH_SH"
- SQL Statement 2
 - Executed 1 time
 - Caused 10 minutes of wait time for end user
 - Waited 90% on "LCK_M_X"

- Where would you begin to improve this?

```
SELECT Top 50 T_ID, T_DTS, ST_NAME,  
TT_NAME, T_S_SYMB, T_QTY, T_EXEC_NAME,  
T_CHRG, S_NAME, EX_NAME  
FROM E_TRADE, E_STATUS_TYPE,  
E_TRADE_TYPE, E_SECURITY, E_EXCHANGE  
WHERE T_CA_ID = 490  
AND ST_ID = T_ST_ID  
AND TT_ID = T_TT_ID  
AND S_SYMB = T_S_SYMB  
AND EX_ID = S_EX_ID  
ORDER BY T_DTS desc;
```

- SQL Diagramming
- Dan Tow
- <http://www.singingsql.com/index.html>



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3. Find all filters
 - a) WHERE clause
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4. Calculate the selectivity

Customers	Orders	Details
19k rows	31k rows	121k rows
Need 18k rows	Need 12k rows	Need 3k rows
0.947 selectivity	0.387 selectivity	0.024 selectivity

Next Steps:

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6. Gather info on existing keys and indexes
7. Examine the execution plan
 - a) SET STATISTICS IO ON
 - b) SET STATISTICS TIME ON
8. Record results

Demo script:

```
SELECT c.CustomerID, soh.ShipDate
FROM Sales.SalesOrderDetail sod
INNER JOIN Sales.SalesOrderHeader soh ON
    sod.SalesOrderID = soh.SalesOrderID
INNER JOIN Sales.Customer c ON c.CustomerID =
    soh.CustomerID
WHERE sod.SpecialOfferID = 2
AND soh.ShipDate between '2003-01-01 00:00:00.000' and
'2004-01-01 00:00:00.000'
AND c.CustomerType = 'I'
```

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 10. Re-run query and examine results and execution plans
 11. Repeat on next lowest selectivity table
 12. Continue on, reducing your logical and physical reads
- WARNING:** Adding additional indexes is *not always* the right thing to do!
- Compare volume of other DUI statements!*

- Run the same measures again!



Now What?



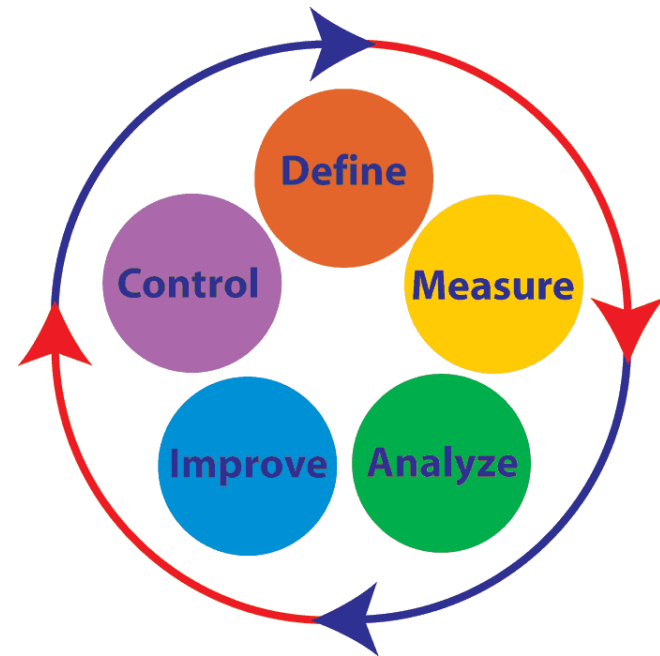
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 - Create “Top Ten” lists
 - Provide weekly summaries

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 - Note significant improvements
 - Create “Top Ten” lists
 - Provide weekly summaries
- Note success for high-profile projects
 - Ask for a raise/promotion/training
 - Explain “what do you do here?”



What Are You Waiting For?

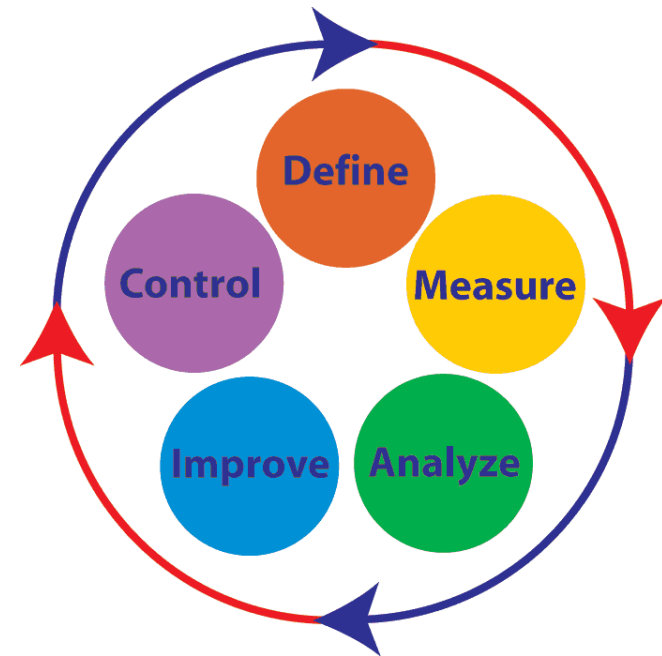
- Get your definitions





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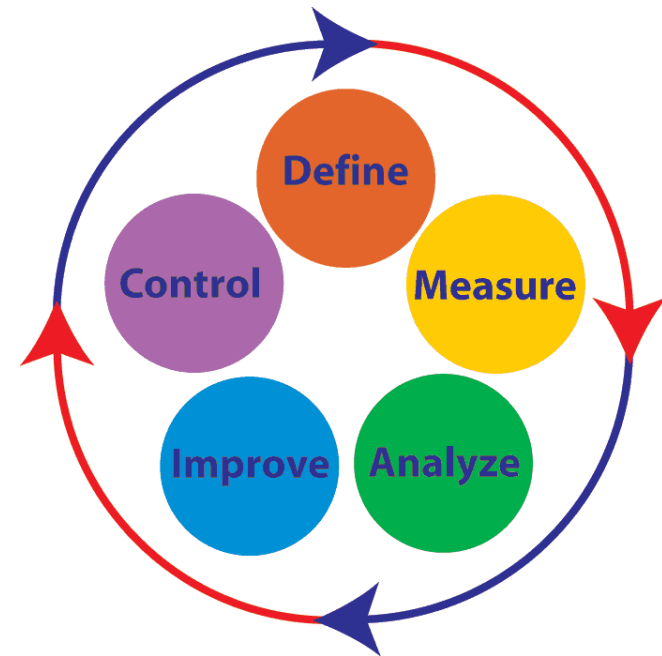
- Get your definitions
- Get your measures in place





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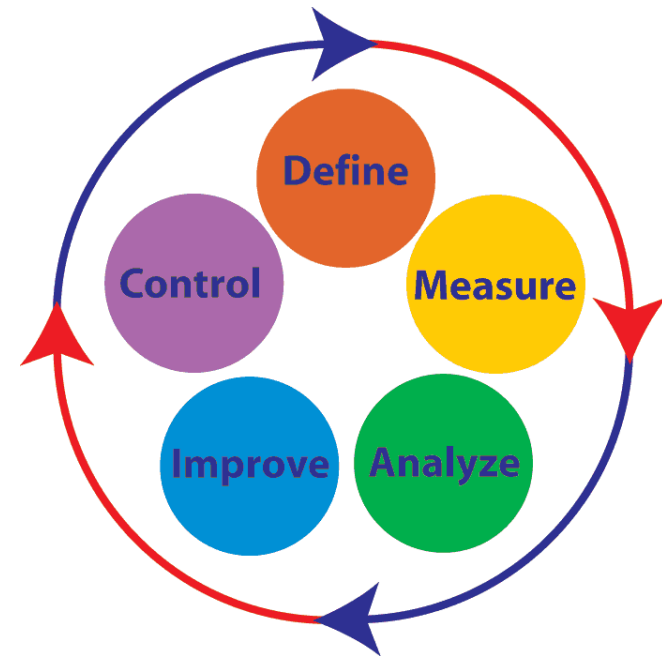
- Get your definitions
- Get your measures in place
- Start analyzing





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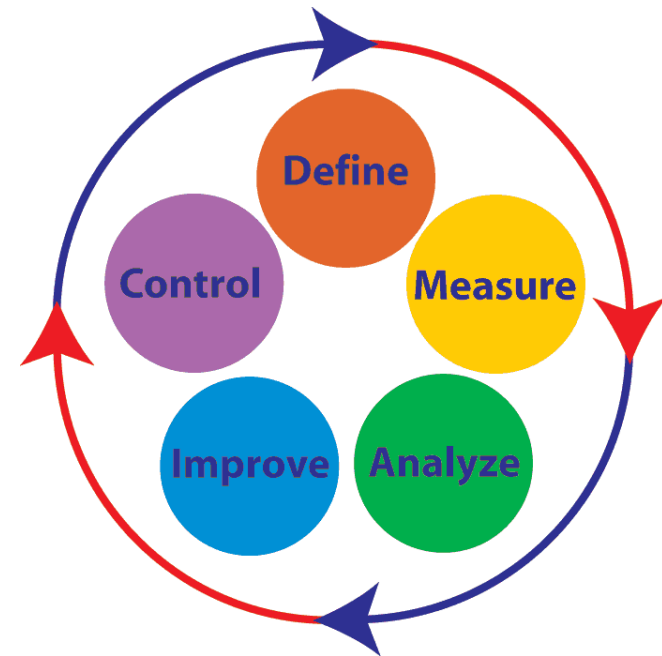
- Get your definitions
- Get your measures in place
- Start analyzing
- Suggest improvements





What Are You Waiting For?

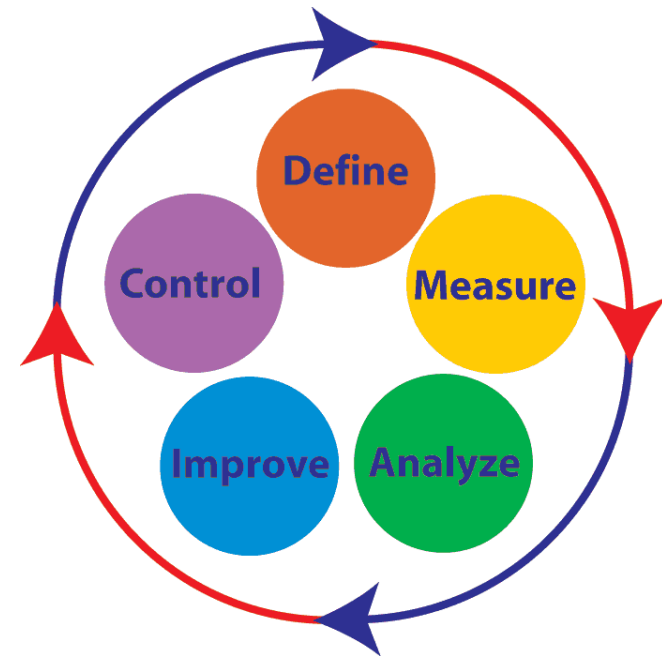
- Get your definitions
- Get your measures in place
- Start analyzing
- Suggest improvements
- Control reports





What Are You Waiting For?

- Get your definitions
- Get your measures in place
- Start analyzing
- Suggest improvements
- Control reports
- Rinse, lather, repeat



For More Information

- <http://technet.microsoft.com/en-us/library/cc966413.aspx>
- <http://pal.codeplex.com/>
- <http://speakerrate.com/speakers/3000-sqlrockstar>
- <http://thomaslarock/presentations>

THANK YOU LOUISVILLE!
NEXT STOP: CLEVELAND!

Questions?

